

Fair, Non-Progressive Influence Control Under Backfire and Parameter Uncertainty

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Abstract—The classical influence maximization guarantee of Kempe, Kleinberg, and Tardos rests on four assumptions: progressive diffusion, a monotone submodular objective, known influence parameters, and no fairness requirement. Real deployments routinely violate all four at once: influence decays and can be actively reversed (backfire, or negative word-of-mouth), edge probabilities are never known exactly, and social interventions must not neglect a disadvantaged subpopulation. This paper studies the combined relaxation. We prove that a sandwich approximation built from two monotone-submodular surrogates yields a $\rho(1 - 1/e)$ guarantee for *every* backfire intensity $\beta \in [0, 1)$, degenerating gracefully to the tight $1 - 1/e$ bound at $\beta = 0$ and admitting a per-instance, directly computable certificate. We derive a closed-form Lagrangian index for the resulting 2-state/3-action per-node control problem, replacing a bisection-plus-value-iteration inner loop with an $O(Km + n \log n)$ -per-round procedure that we measure to be roughly two orders of magnitude faster than the bisection oracle it replaces. We show that $(1 - 1/e)$ -hardness survives backfire exactly, and we state – without overclaiming – a conditional, not-yet-proven result suggesting a further constant-factor worsening for a plausible but unverified property of Feige’s hard instances. Finally, we identify and correct a population-weighting pathology in an existing isoelastic-welfare fairness objective that systematically under-protects small groups, and replace it with an egalitarian reweighting. We implement all of this in a control policy, MF-BWI-Fair, that combines mean-field belief propagation, Bayesian parameter tracking, the closed-form index, and the corrected fairness reweighting. Preliminary experiments on a synthetic stochastic block model benchmark show MF-BWI-Fair sustaining spread under backfire and recovery where one-shot baselines – including independently reimplemented published algorithms (IMM, He–Kempe’s robust Saturate Greedy) – degrade sharply, and show the egalitarian fix raising worst-off-group reach at negligible spread cost. A full multi-graph validation and a benchmark against the original IMM implementation are in progress and are not reported here.

Index Terms—Influence maximization, submodular optimization, restless bandits, fairness, robust optimization, social networks.

I. INTRODUCTION

Kempe, Kleinberg, and Tardos [1] showed that under the Independent Cascade (IC) and Linear Threshold (LT) models, the expected spread of a seed set is monotone and submodular, so the greedy algorithm achieves a $(1 - 1/e)$ -approximation via the general result of Nemhauser, Wolsey, and Fisher [2], and this ratio is tight under standard complexity assumptions [3]. That guarantee, and the large body of scalable algorithms it spawned – CELF’s lazy-forward speedup [4] and the near-linear-time RIS/martingale algorithm IMM [5], [6]

chief among them – rests on four assumptions taken together: (A1) diffusion is *progressive* (once active, always active); (A2) the spread function is *monotone submodular*; (A3) influence parameters are *known* exactly; and (A4) the objective has *no fairness structure* – every node’s activation counts equally toward one aggregate number.

Each assumption is routinely false in the settings influence maximization is used to model. Interventions decay: a vaccinated or informed individual can relapse or forget, and messaging campaigns require sustained reinforcement, not a one-time seeding. Influence can be actively *negative*: word of mouth can backfire, and an over-exposed contact can turn an active neighbor against the message rather than merely fail to convert it [7], [8], [9]. Edge probabilities are estimated from noisy or incomplete cascade data, never known exactly. And in the social-intervention settings that motivate much of this work in the first place – HIV-prevention and stable-housing information campaigns among homeless youth, the application context Rahmattalabi et al. build their fairness-aware formulation around [10] – treating every node as interchangeable is not a simplification but a policy failure: a mechanism that maximizes aggregate reach while systematically under-serving one subpopulation is not simply “unfair” in the abstract, it directly undermines the deployment’s stated public-health goal.

Prior work has relaxed these assumptions one, or occasionally two, at a time (Section II). This paper studies all four relaxations *together*: a model with recovery and backfire-driven deactivation (relaxing A1 and, as we show, A2), Bayesian tracking of unknown influence and recovery parameters (relaxing A3), and an egalitarian-welfare fairness objective over groups (relaxing A4). Our literature search found no existing paper that combines all four axes; the two closest works each combine exactly two (Section II).

Contributions

- **A formal layered-semantics model** (Section III) in which an active node survives a round unless it spontaneously recovers or is backfired by an active in-neighbor, and any node – active or not – can be (re-)activated by a successful positive trial from an active in-neighbor.
- **A proven submodularity result for recovery alone** (Lemma 2): spontaneous, state-independent recovery preserves monotone submodularity of cumulative spread on any graph, via a layered live-edge-graph reachability argument. This isolates recovery as the “free” relaxation; neighbor-dependent backfire is what destroys the coverage structure.

- **A sandwich approximation theorem** (Theorem 1, *proven*, instantiating the general sandwich technique of [11] together with [2]): for *every* backfire intensity $\beta \in [0, 1]$, a greedy solution taken over two monotone-submodular surrogates and the raw objective achieves $\rho(1 - 1/e) \text{OPT}_\beta$, where ρ is an instance-dependent sandwich ratio that degenerates to the tight $1 - 1/e$ bound at $\beta = 0$. The guarantee is also directly instance-certifiable: a returned set ships its own approximation ratio, computable by simulating the two $\beta = 0$ surrogates.
- **A closed-form Lagrangian index** (Propositions 2–4, Theorem 2, *proven* except for indexability itself, which is checked numerically and explicitly flagged as such) for the resulting 2-state/3-action per-node control problem, reducing the per-round decision from bisection with per-node value iteration to $O(Km + n \log n)$ – an approximately $130\times$ speedup measured in this project’s implementation.
- **Hardness results that preserve honest status labels:** Theorem 3 *proves* that $(1 - 1/e)$ -inapproximability survives backfire exactly, for every fixed β ; Proposition 5 is stated and discussed as a *conditional, unproven* result – consistent with its labeling in the underlying project notes – suggesting (not proving) a further constant-factor worsening contingent on one plausible but unverified property of Feige’s hard-instance construction.
- **A corrected fairness mechanism:** we identify a measured population-weighting pathology in the isoelastic welfare objective of [10] (small groups are protected only once under-served by a factor proportional to a group-size ratio) and replace it with an egalitarian (population-unweighted) reweighting that protects the worst-off group at a neutral default setting.
- **MF-BWI-Fair**, a control policy combining mean-field belief propagation, Bayesian posterior tracking, the closed-form Lagrangian index, and the corrected welfare reweighting, together with a preliminary comparison against six algorithms including independently-verified reimplementations of two published baselines.

II. RELATED WORK

A. Classical influence maximization

Kempe, Kleinberg, and Tardos [1] established monotone submodularity of expected spread under IC/LT and the resulting $(1 - 1/e)$ greedy guarantee [2], tight under $P \neq NP$ [3]. CELF [4] gives a lazy-forward speedup exploiting the same submodularity. IMM [5], building on the RIS/reverse-reachable-set duality of [6], gives an $O((k+\ell)(n+m) \log n/\epsilon^2)$ near-linear-time $(1 - 1/e - \epsilon)$ -approximation and is, to our knowledge, the most-cited scalable classical algorithm with publicly available original code [12].

B. Fair / group-fairness-constrained influence maximization

Tsang et al. [13] introduce maximin and diversity-constraint fairness notions as a multi-objective submodular problem. Fish et al. [14] study a maximin welfare objective directly and prove it is not monotone submodular in general. Farnadi

et al. [15] unify several fairness notions as exact MILP formulations at small scale, with no submodular approximation guarantee. Becker et al. [16] show deterministic maximin-fair seeding is hard to approximate but that randomized seeding recovers a $(1 - 1/e - \epsilon)$ -type guarantee. Rahmat-talabi et al. [10] propose an isoelastic (CES) welfare objective over per-group reach and prove monotone submodularity via the concave-nondecreasing-composed-with-monotone-submodular lemma of [17], giving an exact $(1 - 1/e)$ guarantee – but only for the classical progressive IC model; their proof does not address deactivation or negative influence. This paper adopts their objective *form*, not their guarantee, and additionally corrects a population-weighting side effect of their exact formula (Section V-A). Rui et al. [18] and Feng et al. [19] extend fairness to scalable and data-driven/learning-based settings, respectively; the latter is out of scope for the classical-algorithms focus here.

C. Non-progressive / recovery-capable influence maximization

Chan and Ning [20] show that non-progressive dynamics under the LT model lose submodularity on general graphs, recovering it only on DAGs. Golnari et al. [21] propose the “Heat Conduction” model, proving submodularity under a specially engineered non-progressive rule. Lou et al. [22] give a continuous-time two-state Markov estimation model without a submodularity result. Zahoor et al. [23] and Hui et al. [24] study reactivation dynamics, the latter via deep reinforcement learning (out of scope here). Taninmis et al. [25] study an adversarial deactivation game, a different framing from the spontaneous, non-adversarial backfire studied here. Our Lemma 2 gives a stronger/different result than Chan and Ning’s DAG-restricted finding: spontaneous, state-independent recovery alone preserves monotone submodularity on *any* graph, via layered-DAG reachability rather than threshold-rule structure.

D. Non-monotone / non-submodular / negative-influence models

Bharathi et al. [26] give the first competitive cascade model. Chen et al. [7] extend IC with a quality factor allowing activated nodes to flip negative, engineered to remain submodular. He et al. [8] introduce a two-competing-cascade “sandwich approximation” as a scalability device, whose general form is due to Lu, Chen, and Lakshmanan [11]; our Theorem 1 instantiates the latter’s general monotone-submodular bracket $L \leq f \leq U$, applied to a materially different bracket – a recovery-only process, not a two-cascade one. Li et al. [9] give a signed-network model (IC-P) that also remains submodular by construction. Buchbinder et al. [27] and Tukan et al. [28] give tight and practical constant-factor guarantees for unconstrained/cardinality-constrained *non-monotone submodular* maximization – a different relaxation axis (non-monotonicity of a still-submodular function) from ours. Bian et al. [29] give a submodularity-ratio/curvature-based greedy guarantee for non-submodular functions, superseded here by the sandwich approach, which holds for all β rather than requiring an estimated ratio. Horel and Singer [30] give an

exponential query-complexity lower bound for functions ε -far from submodular once $\varepsilon = \omega(n^{-1/2})$, but only for an *adversarial* perturbation; our backfire perturbation is structured (a bounded extra kill rate), which is why Theorem 1 escapes this limitation (Section IV-D).

E. Bayesian, robust, and online influence maximization under uncertainty

He and Kempe [31] formalize worst-case-across-scenarios robust influence maximization and give a bicriteria Saturate Greedy algorithm with a provable budget-blowup/quality trade-off; no official code is available for this specific paper. The dominant paradigm for online uncertainty is combinatorial-bandit learning [32], [33], [34], [35], [36], [37] or one-shot distributionally robust optimization [38], [39]. A genuinely single-shot Bayesian formulation – integrating expected spread over a prior without adaptivity or worst-case hedging – exists only embedded inside bandit posterior-update machinery, not as a standalone anchor result; this is the gap our Bayesian tracker sits in.

F. Restless-bandit sequential control

Mate et al. [40] give binary-action Whittle-index results for collapsing bandits, explicitly leaving multi-action indexability future work. Killian, Perrault, and Tambe [41] give the Lagrangian-relaxation template for general multi-action restless bandits that our closed-form index specializes and simplifies for the specific 2-state/3-action case, avoiding the general indexability problem they call “notoriously difficult.” Ou et al. [42] propose an alternative to mean-field network coupling that we verified does not apply here: their coupling is through a deterministic, known action-vector, not through another arm’s actual random state, which is the genuinely bilinear coupling our setting requires. Li and Varakantham [43] study fairness constraints in independent-arm restless bandits, confirming fairness-in-RMAB as an active but separate line of work.

G. Positioning

No paper we found combines group fairness, non-progressive/recovery dynamics, non-monotone backfire influence, and Bayesian parameter uncertainty. The two closest works each combine exactly two of these four axes, both very recent: a 2026 PACMMOD paper combining robust fairness with multiple-community-partition (parameter) uncertainty [44], and Fang et al.’s 2026 scalable fair influence *blocking* maximization, combining fairness with non-monotone/blocking-style negative influence [45]. Separately, no paper we found connects submodular/RIS-style influence-maximization theory to restless-bandit sequential control theory at all; Section IV-C is, to the depth of our search, the first to do so for a node-level activation/maintenance control problem.

III. PROBLEM FORMULATION

Let $G = (V, E)$ be a directed graph, $n = |V|$, $m = |E|$, with a partition of V into groups $G_1, \dots, G_M$. Rounds are $t = 0, 1, \dots, T$; the active set $A_t \subseteq V$ starts at $A_0 = S$, the seed set. Each round draws independent randomness: for every edge (u, v) , a positive trial $X_{uv}^t \sim \text{Bern}(p_{uv})$ and a backfire trial $Y_{uv}^t \sim \text{Bern}(\beta p_{uv})$, for a global backfire intensity $\beta \in [0, 1]$; for every node v , a recovery trial $R_v^t \sim \text{Bern}(q_v)$.

Transition (layered semantics). $v \in A_{t+1}$ iff

$$\begin{aligned} &[v \in A_t \wedge R_v^t = 0 \wedge \forall u \in A_t \cap N_{\text{in}}(v) : Y_{uv}^t = 0] \\ &\vee [\exists u \in A_t \cap N_{\text{in}}(v) : X_{uv}^t = 1]. \end{aligned} \quad (1)$$

An active node stays active unless it recovers or is backfired by an active in-neighbor, and any node – active or not – is (re-)activated by a successful positive trial from an active in-neighbor; in particular a positive trial can *rescue* an active node in the same round it would otherwise have recovered or been backfired. This rescue clause is exactly what makes the one-step map $A_t \mapsto A_{t+1}$ monotone in A_t at $\beta = 0$, which the submodularity result of Section IV rests on; for $\beta > 0$ the map is not monotone, since the backfire clause is anti-monotone.

The objective is cumulative expected spread,

$$f_\beta(S) = \mathbb{E} \left[\sum_{t=1}^T |A_t| \right], \quad A_0 = S, \quad (2)$$

time-averaged spread being the same quantity up to the constant $1/T$.

Fairness. Let u_g denote group g ’s time-averaged realized reach fraction (active fraction of G_g , averaged over rounds). Following an isoelastic (constant elasticity of substitution) welfare form [10], define

$$W_\alpha(u) = \sum_g N_g \frac{u_g^\alpha}{\alpha} \quad (\alpha \neq 0), \quad W_0(u) = \sum_g N_g \log u_g, \quad (3)$$

population-weighted by group size N_g ; $\alpha \rightarrow 1$ is utilitarian, $\alpha = 0$ is proportional fairness, and $\alpha \rightarrow -\infty$ approaches leximin. As detailed in Section V-A, this project identifies and corrects a measured pathology in the population-weighted form and adopts, by default, the egalitarian (population-unweighted) variant $W_\alpha(u) = \sum_g u_g^\alpha / \alpha$ instead, which counts each group once regardless of size.

IV. THEORETICAL RESULTS

Statements below preserve the status labels of the underlying project notes exactly: *proven* (theorem, proposition, lemma, with a proof or complete proof sketch), *cited* (an external result relied upon), or *conjectural/conditional* (stated precisely, with the reason it is not yet proven). No conjectural or conditional statement is used as a premise for any theorem.

A. Submodularity of the recovery-only process

Lemma 1 (Layered live-edge representation). *Fix $\beta = 0$. Build a layered graph $\mathcal{L}$ on $V \times \{0, \dots, T\}$ with, for each round t , a self-edge $(v, t) \rightarrow (v, t+1)$ live iff $R_v^t = 0$, and a cross-edge $(u, t) \rightarrow (v, t+1)$ for each $(u, v) \in E$, live*

iff $X_{uv}^t = 1$. Then for every realization of the randomness, $A_t = \{v : (v, t) \text{ reachable from } S \times \{0\} \text{ along live edges}\}$.

Proof. By induction on t ; at $t = 0$ both sides equal S , and the inductive step is exactly the layered-semantics transition rule read as a reachability statement. ■

Lemma 2 (Recovery alone preserves monotone submodularity). *For $\beta = 0$ and any recovery rates $q_v \in [0, 1]$, $f_0(S)$ is monotone and submodular in S .*

Proof. For each fixed realization, $|A_t|$ is the number of layer- t nodes reachable from $S \times \{0\}$ in a fixed DAG (Lemma 1) – a coverage function of S , hence monotone submodular; sums over t and expectations over realizations of monotone submodular functions remain monotone submodular. ■

This isolates exactly which relaxation is "free": spontaneous, state-independent recovery is harmless (it is a random self-edge and reachability stays a coverage function), while neighbor-dependent deactivation (backfire) destroys the coverage structure, since an active in-neighbor can now *remove* a node from the reachable set. This sharpens the picture in Chan and Ning [20], who lose submodularity under non-progressive *threshold* rules where deactivation depends on neighbor states; recovery alone needs no such structure to stay submodular, on any graph.

B. A sandwich approximation for all backfire intensities

Let $U(S) := f_0(S)$ (the $\beta = 0$ process at the true recovery rates). For node v , let $\delta_v := 1 - \prod_{u \in N_{\text{in}}(v)} (1 - \beta p_{uv}) \leq \beta \Delta_{\text{in}}$, where $\Delta_{\text{in}} := \max_v \sum_{u \in N_{\text{in}}(v)} p_{uv}$, and let $L(S)$ be the $\beta = 0$ process under inflated recovery $q'_v := 1 - (1 - q_v)(1 - \delta_v)$.

Lemma 3 (Upper coupling). *Under shared randomness, $A_t^\beta \subseteq A_t^0$ for all t ; hence $f_\beta(S) \leq U(S)$.*

Lemma 4 (Lower coupling). *There is a coupling under which $A_t^L \subseteq A_t^\beta$ for all t ; hence $L(S) \leq f_\beta(S)$.*

Both couplings are proven by induction, replacing the per-edge backfire trials with a single uniform draw compared against the aggregate kill probability and, respectively, against δ_v ; the key inequality is that β -kill implies L -extra-recovery because the β -kill threshold is at most δ_v (attained when every in-neighbor is active).

Theorem 1 (Sandwich approximation, all β). *L and U are monotone submodular (Lemma 2) and $L \leq f_\beta \leq U$ pointwise (Lemmas 3–4). Let S_L, S_U be the greedy solutions for L, U under $|S| \leq k$, S_β the greedy solution run directly on f_β , and $S_{\text{sand}} = \arg \max_{S \in \{S_L, S_U, S_\beta\}} f_\beta(S)$. Then, with S^* optimal for f_β ,*

$$f_\beta(S_{\text{sand}}) \geq \max \left\{ \frac{f_\beta(S_U)}{U(S_U)}, \frac{L(S^*)}{f_\beta(S^*)} \right\} \left(1 - \frac{1}{e}\right) f_\beta(S^*) \geq \rho \left(1 - \frac{1}{e}\right) f_\beta(S^*) \quad (4)$$

where $\rho := \min_{|S| \leq k} L(S)/U(S)$.

Proof. Cited: the sandwich-approximation theorem of Lu, Chen, and Lakshmanan [11], applied to the bracket $L \leq f_\beta \leq U$; the $(1 - 1/e)$ factors are [2] applied to greedy on

each submodular surrogate. The final inequality follows from $L(S^*)/f_\beta(S^*) \geq L(S^*)/U(S^*) \geq \rho$. ■

Theorem 1 improves on treating backfire as an arbitrary perturbation from submodularity via [30], which would only hold for $\beta = O(n^{-1/2})$ – far below the $\beta \in [0.1, 0.7]$ regime our empirical study targets – because it exploits the *structure* of the perturbation (a bounded extra kill rate) rather than treating it adversarially, and it holds for every β , recovering the tight $(1 - 1/e)$ exactly at $\beta = 0$ (where $L = U$, $\rho = 1$). It also yields a practical corollary: since L and U are ordinary $\beta = 0$ processes that can be simulated directly, $f_\beta(S)/U(S)$ is a directly computable, per-instance certificate, $f_\beta(S) \geq \frac{f_\beta(S)}{U(S)} (1 - \frac{1}{e}) \text{OPT}_\beta$ – the algorithm ships its own approximation guarantee on every instance it is run on.

Proposition 1 (Finite-horizon bound on ρ). $\rho \geq (1 - \beta \Delta_{\text{in}})^T$.

This is proven by charging every self-edge of a witness reachability path under the coupling of Lemma 4, and is conservative (it ignores re-activation through alternate paths). A tighter steady-state conjecture, $\rho \gtrsim q/(q + \beta \Delta_{\text{in}})$ for $q = \min_v q_v$, is stated in the underlying project notes as an explicit *conjecture*: the heuristic single-sojourn argument does not account for the downstream cascade a shortened sojourn fails to seed, and closing that gap is left open. This conjecture is not used in any proof in this paper (Section VIII discusses why it is, in any case, not the most promising target for tightening the sandwich ratio).

C. A closed-form index for the control problem

Given the mean-field belief field (Section V), each node's per-round control problem is a 2-state MDP – states $s \in \{0, 1\}$, discount γ , reward w at $s = 1$ – with natural transitions $a = p_{01}$ (activate), $b = p_{10}$ (deactivate), and three actions: NONE (free, either state), CONVERT (cost c_C , forces $0 \rightarrow 1$, only at $s = 0$), MAINTAIN (cost c_M , forces stay-at-1, only at $s = 1$), under a shared Lagrange price λ per unit cost. There are exactly four deterministic stationary policies, named by (action at 0, action at 1): NN, NM, CN, CM.

Proposition 2 (Closed forms). *With $D := (1 - \gamma)(1 - \gamma + \gamma(a + b))$, each policy's value functions $V_1(\lambda), V_0(\lambda)$ are given in closed, affine-in- λ form by solving its fixed 2×2 linear system (e.g., for CN, $V_0 = -\lambda c_C + \gamma V_1$ and $V_1 = w + \gamma[(1 - b)V_1 + bV_0]$, giving $V_1 = (w - \gamma b \lambda c_C)/((1 - \gamma)(1 + \gamma b))$); the full table of four closed forms is given in the underlying project notes and reproduced in the algorithm's implementation.*

Proposition 3 (Convexity in λ). $V^*(s, \lambda) = \max_\pi V_\pi(s, \lambda)$ over the four policies is a maximum of four affine functions of λ : convex, piecewise linear, with at most three breakpoints, computable in $O(1)$.

Proof. In a discounted finite MDP some stationary deterministic policy is optimal at every state simultaneously, so V^* is exactly this finite max; Killian, Perrault, and Tambe [41] (their Prop. 4.1) prove convexity in λ for the general multi-action restless-bandit case, of which this is a one-line special case via finite enumeration of the four policies. ■

Proposition 4 (Index and indexability, this case). *Define node v 's index λ_v^* as the largest $\lambda \geq 0$ at which the optimal policy still takes the paid action at v 's current state.*

A proof *sketch* shows the paid action's advantage over NONE has a strictly negative slope in λ from the cost term, while the value gap $V_1^* - V_0^*$ is non-increasing in λ on each linear piece, checked by finite case analysis over the four policies. **We state plainly, matching the underlying project notes' own status label, that indexability – i.e., that the paid set is a down-set $[0, \lambda_v^*]$ – is only verified numerically here** (on random instances, over a fine λ grid, checking that per-node cost is non-increasing in λ), not proven in general; the finite four-policy structure makes this tractable to check exactly where Killian et al. [41] call general multi-action indexability "notoriously difficult" and avoid it. The index itself is defined as the supremum of the paid set, which remains well-defined and exactly computable regardless.

Theorem 2 (Per-round complexity). *Given Propositions 2–4, one round of the control policy costs $O(Km + n \log n)$, where K is the number of mean-field fixed-point iterations (geometric convergence under a contraction condition on $\Delta \max(p^+, p^-)$).*

The $n \log n$ term sorts nodes by λ_v^* and fills the shared budget greedily – exactly "smallest λ with aggregate cost $\leq B$," the target of the bisection-plus-value-iteration procedure it replaces, which cost $O(n \cdot I_{\text{bisection}} \cdot I_{\text{VI}})$ (on the order of 8000 n scalar operations per round in our configuration); this project's own measurements put the closed-form replacement at roughly 130 $\times$ faster per round than that bisection oracle. For comparison, naive Monte Carlo KKT greedy costs $O(knRm)$ per seed set, and one static IMM selection costs $O((k+\ell)(n+m) \log n/\varepsilon^2)$ [5]: our per-round cost is near-linear like a single IMM run but is paid T times, the stated price of sustaining influence under decay rather than seeding once.

D. Hardness

For $\beta = 0$ (and $q = 0$, large T), the problem contains max- k -cover, so $1 - 1/e$ is an unbeatable ceiling for every $\beta \geq 0$ under $P \neq \text{NP}$ [3], [2] – adding backfire cannot remove hard instances on which β is simply irrelevant. For adversarial, arbitrary non-submodular perturbations, no constant-factor approximation exists in general once a function is $\omega(n^{-1/2})$ -far from submodular [30]; Theorem 1 escapes this only because our perturbation is structured, not adversarial.

Theorem 3. *For every fixed $\beta \in [0, 1)$, no polynomial-time algorithm approximates f_β within $(1 - 1/e + \varepsilon)$ for any $\varepsilon > 0$ unless $P = \text{NP}$.*

Proof. Reduction from max- k -cover, $(1 - 1/e + \varepsilon)$ -inapproximable unless $P = \text{NP}$ [3]. Build a bipartite digraph (set-nodes to element-nodes, edge probability 1), each element replicated to swamp the additive seed-count term. The key subtlety is that backfire is realized on this instance (a seed keeps attacking every element it covers), so its harmlessness must be shown, not assumed. On this instance class, an

element with $c \geq 1$ active covering seeds throughout is a two-state chain with up-rate 1 and down-rate $1 - (1 - \beta)^c$, giving an exact value formula $f_\beta(S) = kT + T \sum_v \varphi_\beta(c_v(S)) + O(n)$ with $\varphi_\beta(c) = 1/(2 - (1 - \beta)^c)$ strictly decreasing in $c \geq 1$ and bounded below by $1/2$; the max-coverage gap therefore survives backfire intact. ■

This value formula also shows $f_\beta \geq f_0/2$ for every seed set on this instance class at $q = 0$, exactly where the sandwich ratio ρ of Theorem 1 is vacuous ($\rho = 0$).

Proposition 5 (Conditional – not a theorem). *Suppose Feige's NO-instances additionally satisfy: for every candidate k -subset S , the number of elements covered exactly once is at most $(1/e + \varepsilon)n$. Then no polynomial-time algorithm achieves ratio $c(\beta)(1 - 1/e) + \varepsilon'$, where*

$$c(\beta)(1 - 1/e) = \frac{1}{e} + \left(1 - \frac{2}{e}\right) \frac{\varphi_\beta(2)}{\varphi_\beta(1)} = (1 - 1/e) - \left(1 - \frac{2}{e}\right) \frac{\beta(1 - \beta)}{1 + 2\beta - \beta^2}. \quad (5)$$

Numerically $c(\beta) \approx 0.968, 0.946, 0.939, 0.940, 0.960, 0.981$ at $\beta = 0.1, 0.25, 0.4, 0.5, 0.75, 0.9$, worst near $\beta \approx 0.45$, vanishing at both ends. **We flag this explicitly, and only, as a conditional result, exactly as it is labeled in the underlying project notes:** the extra property is not part of Feige's stated result, though it is plausible (a Chernoff/union-bound concentration argument for the "covered-exactly-once" statistic under a random partition), and two gaps remain – (i) the analogous block-averaging step needs concavity of $d \mapsto \mathbb{E} \varphi_\beta(\text{Bin}(d, 1/k))$, unchecked; (ii) cross-terms from two sets in the same partition class need the reduction's pairwise-soundness property re-verified for this statistic. We report this as "very likely true, not yet proven," and do not use it as a premise for any theorem in this paper. Structural evidence – a per-edge revive-at-least-as-fast-as-kill asymmetry, and several natural gadget constructions all failing for the same underlying reason – suggests no super-constant worsening is achievable for fixed β , but this too is offered as evidence, not a proof.

V. ALGORITHM: MF-BWI-FAIR

MF-BWI-Fair (Mean-Field Belief with a Lagrangian index and isoelastic-Welfare reweighting, Fair) is a stateful per-round control policy. Each round:

- 1) **Bayesian posterior tracking.** A Beta-Bernoulli tracker maintains posterior means $\hat{p}_{uv}^+, \hat{p}_{uv}^-, \hat{q}_v$ for every edge/node from observed trial outcomes; the policy never sees the simulator's hidden true parameters, only these running estimates.
- 2) **Mean-field belief fixed point.** Exact per-node marginals require tracking correlations across the whole active-set distribution, intractable at scale. We use the standard individual-based / N-intertwined mean-field approximation from network epidemic models: each node's in-neighbors are replaced by their marginal activation prob-

Algorithm 1 MF-BWI-Fair, one round

Require: graph G , state s , budget B , fairness exponent α

- 1: $(\hat{p}^+, \hat{p}^-, \hat{q}) \leftarrow$ posterior means from tracker
- 2: $m \leftarrow$ mean-field fixed point of Eq. (5) from $(s, \hat{p}^+, \hat{p}^-, \hat{q})$
- 3: $(p_{01}, p_{10}) \leftarrow$ field transition probabilities from m
- 4: $u \leftarrow$ tracked per-group time-averaged reach
- 5: $w \leftarrow \text{GROUPWELFAREWEIGHTS}(u, \alpha)$ (egalitarian default)
- 6: **for** each node v **do**
- 7: build 2-state/3-action MDP with reward $w_{g(v)} \cdot s_v$, kernel $(p_{01}(v), p_{10}(v))$
- 8: $\lambda_v^* \leftarrow$ closed-form index (Prop. 4)
- 9: **end for**
- 10: sort nodes by λ_v^* descending; select prefix fitting in B
- 11: greedily fill leftover budget by descending index among unselected nodes
- 12: apply CONVERT/MAINTAIN to selected nodes, NONE elsewhere
- 13: observe transition, update Bayesian tracker and u_g

abilities m_u , giving the self-consistency equation

$$m_v = m_v(1 - \hat{q}_v) \prod_{u \in N_{\text{in}}(v)} (1 - \hat{p}_{uv}^- m_u) + (1 - m_v) \left(1 - \prod_{u \in N_{\text{in}}(v)} (1 - \hat{p}_{uv}^+ m_u) \right), \quad (6)$$

solved by Jacobi fixed-point iteration to tolerance 10^{-4} or 50 iterations. We verified directly that the alternative network-coupling method of Ou et al. [42] does not apply here, since it requires a deterministic action-vector coupling rather than the genuinely bilinear neighbor-state coupling this model has.

- 3) **Group welfare weights.** Each group's current marginal welfare weight w_g is computed from its tracked time-averaged realized reach u_g (Section V-A), egalitarian by default.
- 4) **Closed-form Lagrangian index.** Using the converged belief field to obtain each node's "no-action" transition probabilities $p_{01}(v), p_{10}(v)$, each node's 2-state/3-action MDP is built with reward $r_v(s) = w_{g(v)} \cdot s$, and the shared-budget Lagrangian relaxation is solved via the closed-form index of Section IV-C (a bisection-plus-value-iteration oracle is kept as a test reference).
- 5) **Leftover-budget fill.** Any budget left over from the index scan's discreteness is filled by a greedy tie-break on each node's closed-form index (or Bellman gap, under the bisection oracle), keeping the hard budget constraint intact without reintroducing per-group floors or quotas.
- 6) **Execute, observe, update.** Actions are applied; the simulator advances one round under the layered-semantics transition rule (Section III); observed trial outcomes update the Bayesian tracker; and each group's realized reach average u_g is updated for the next round's weights.

A. The fairness fix: population-weighted vs. egalitarian welfare

Rahmattalabi et al.'s isoelastic welfare weight, implemented exactly as published, is $w_g = N_g u_g^{\alpha-1}$ (or N_g/u_g at $\alpha = 0$): the correct marginal utility for maximizing *aggregate* welfare, in which a group's contribution scales with its population. This has a concrete, measured downside for the stated goal of not neglecting a small group: at $\alpha = 0$, group g 's weight exceeds group g' 's only once $u_{g'}/u_g > N_{g'}/N_g$ – the smaller group must be under-served by a factor proportional to the group-size *ratio* before the mechanism reacts at all. On this project's own three-group (20/40/60) benchmark graph, at a neutral $\alpha_{\text{fair}} = 0$ this kept the smallest group at a visibly lower realized reach than the largest purely because a $3\times$ reach gap was required before the weight formula began favoring it. We therefore adopt, as the default, the egalitarian (population-unweighted) form $w_g = u_g^{\alpha-1}$ (or $1/u_g$ at $\alpha = 0$), a standard alternative welfare form in the social-welfare and fair-division literature (not a new mechanism), counting each group once regardless of size – the appropriate choice when fairness means each *group*, not each *person*, is treated fairly regardless of population.

VI. EXPERIMENTAL SETUP

Graph. The primary validation graph is a single synthetic stochastic block model with three groups of unequal size (20/40/60, 120 nodes total), within-block edge probability $p_{\text{in}} = 0.07$, between-block edge probability $p_{\text{out}} = 0.008$, and per-edge positive-influence probability $p_{uv} \sim \text{Uniform}(0.05, 0.15)$. Density was hand-tuned so the network neither saturates near full activation nor fails to cascade at all, either of which would mask the effects under study. **We state plainly that this is the only graph topology validated in this draft;** a multi-graph and real-dataset validation is in progress (experiments/MULTIGRAPH_VALIDATION.md) and is not reported here.

Budget and horizon. Per-round budget $B = 100$, horizon $T = 30$ rounds. One-shot baselines are given $k = 20$ seeds, chosen so that $k \cdot \text{cost}(\text{CONVERT}) = 100 = B$: one round's worth of MF-BWI-Fair's budget, spent once at round 0.

Baselines. *KKT-greedy/CELF* [1], [4]: classical Monte Carlo greedy under plain progressive IC ($\beta = 0, q = 0$), this project's own reimplementation, used as the "ignore all four relaxations" ablation. *Fair-greedy*: an in-house welfare-weighted greedy baseline in the style of [10] maximizing a sum of concave per-group log-utilities. *Repeated-greedy*: an in-house cost-effective greedy that re-selects actions every round under the same per-round budget, using *true* parameters directly (no Bayesian uncertainty, no fairness mechanism) – the strongest plausible classical competitor that also acts repeatedly, included specifically to isolate whether MF-BWI-Fair's advantage is more than "gets to act every round." *IMM* [5]: this project's independent reimplementation of the two-phase martingale sampling and node-selection algorithm, cross-checked against the paper's own pseudocode and against a documented erratum [12]. *He-Kempe Saturate Greedy* [31]: this project's independent reimplementation of the bicriteria

robust algorithm, given the low/high corners of the p_{uv} range as its scenario set. *MF-BWI-Fair*, as described in Section V.

Honesty notes on baseline provenance. Other than the objective forms they target, every baseline in this comparison is an independently-verified reimplementation, not the original authors' code. Official code exists for IMM (a SourceForge project by the original authors) but is not what is benchmarked here; no official code was found for He–Kempe's Saturate Greedy or for Rahmattalabi et al.'s welfare mechanism specifically. **A direct benchmark against IMM's original C++ implementation is in progress** (experiments/IMM_VALIDATION.md) and is not reported here; until it lands, all reported IMM numbers should be read as characterizing this project's own reimplementation, cross-checked against the paper's stated algorithm and theorem statements, not as reproducing the original authors' published numbers.

Metrics. (a) time-averaged active-node count over the full horizon; (b) per-group reach fraction (own-group-size-normalized, time-averaged) and its minimum across groups; (c) wall-clock runtime. Sweeps are conducted over backfire intensity β , recovery rate q , and the fairness exponent α_{fair} , each with 15 independent trials using common random numbers across the swept parameter within a trial to reduce sweep-trend noise without narrowing between-trial variance.

VII. RESULTS

VIII. DISCUSSION AND LIMITATIONS

Single graph topology. All findings to date are demonstrated on one synthetic stochastic block model; generalization across graph families, degree distributions, and real cascade datasets is the subject of ongoing, not-yet-reported work.

The conditional hardness result is exactly that. Proposition 5 is not a theorem: it depends on one plausible but unverified property of Feige's construction, and two identified technical gaps (a concavity step and a cross-term pairwise-soundness check) remain before it could become one. We report the qualitative structural evidence – the revive-at-least-as-fast-as-kill asymmetry, and several gadget constructions all failing to encode a hard-to-decide dependence on the seed set – as suggestive that no super-constant worsening exists for fixed β , not as a substitute for a proof.

No comparison against the original authors' code for He–Kempe or Rahmattalabi et al. Neither paper has published code corresponding to the specific algorithm reimplemented here (a superficially related repository for Rahmattalabi et al. in fact implements a different paper, Tsang et al. [13], and must not be conflated with it). Only IMM has verifiable original code, and even there this draft benchmarks against our own reimplementation pending the validation noted in Section VI.

The mean-field approximation's own error is not separately quantified. The belief fixed point of Section V is an approximation whose deviation from the true (intractable) joint activation distribution has not been bounded or measured in isolation from the end-to-end policy's performance; this is a natural target for future analysis.

The sandwich bound is demonstrably loose, and this is a direction, not a hidden flaw. Exact computation on small instances (2–8 nodes, $T \sim 300$) shows the sandwich ratio ρ loose by up to an order of magnitude where it applies at all (e.g., roughly $13\times$ on a directed star instance, because the lower surrogate inflates every node's recovery uniformly by δ_v even where a node has no active in-neighbor left), and vacuous whenever $q = 0$ or $q + \beta\Delta_{\text{in}} \geq 1$. Separately, f_β/f_0 is not bounded below by any instance-independent constant (a bidirected K_6 instance at $\beta = 0.5, q = 0.05$ collapses by a factor of roughly $300\times$ relative to f_0), so no purely multiplicative comparison against f_0 can be the source of a strong general guarantee. Yet greedy run on the plain $\beta = 0$ surrogate landed within 1% of OPT_β on every one of the six small instances tested – suggestive, not proof, that the practical quality of surrogate-greedy substantially exceeds what $\rho(1-1/e)$ certifies. We read this as the concrete next step for this line of work: a sharper *upper*-bound analysis (a per-instance or instance-class argument in the spirit of the exact layered value formula behind Theorem 3), rather than a search for a matching lower bound on ρ itself, which the evidence here suggests is not the right quantity to target.

Indexability is checked, not proven. As stated in Proposition 4, the down-set property underlying the closed-form index's exactness is verified numerically on random instances over a fine λ grid, not established as a general theorem for this 2-state/3-action case.

An open, unconditional route to a matching lower bound. The underlying project notes identify, but do not pursue to completion, a possible unconditional alternative to Proposition 5: the class of functions $S \mapsto \sum_v \varphi(c_v(S))$ is closed under the same symmetrization used in Vondrák's symmetry-gap framework [46], and the symmetric-instance gap for φ_β (Theorem 3's proof) works out numerically close to, and slightly below, Proposition 5's constant. Whether Vondrák's smoothing-step argument – which uses submodularity directly – extends to this non-submodular but structured class is open and was not checked; if it does, it would give an unconditional oracle-model lower bound without needing the unverified property Proposition 5 depends on. We flag this as the single most promising next step toward turning Section IV-D's conditional result into a proof.

IX. CONCLUSION

We formalized a combined relaxation of the four assumptions behind the classical influence-maximization guarantee – progressive dynamics, monotone submodularity, known parameters, and fairness-free objectives – and proved that a sandwich approximation recovers a $\rho(1-1/e)$ guarantee for every backfire intensity, with a closed-form index making the resulting control problem tractable at near-linear per-round cost. We proved that $(1-1/e)$ -hardness survives backfire exactly, while explicitly declining to upgrade a plausible but unproven constant-factor worsening result to theorem status. We corrected a measured fairness pathology in an existing welfare objective and implemented the result in MF-BWI-Fair. Preliminary single-graph experiments are encouraging;

a full validation against additional graph families and against IMM's original implementation is in progress and left to future revisions of this manuscript.

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